Cloud Computing
Assignment

Name:- Virodula Koushik Sai Raj Roll no: 23010107S

Github:- github.com/viro23010107S-art/cloud-Computing-assignment.

1) Learning)

I had learnt how distillan in knowledge works in practice. A large teacher (phi-3.5) labbeled 3000 commands, and also a student only on teacher's nent. This is seq-level dishillan. The student reached 100% valid JSON & 57.85% full frame accuracy. There is nearly 57% for teachers. Due to this gap I wald n't consider by model is not better in student percepn. In cloud I learnt that obj storage for dataset & models, jobs for parallel work, a deployments with robes, and service and ingress for traffic. I learnt how stateless NACL work, replies need their own rles, and a delay we must have a lower number than that allow it overrides. Finally I learnt to turn measurements into cost per some requests accordingly to the report results.

2) Challenges)

Teacher O/P - My first prompt gave only 60% valid JSON because model kept inventing extra command. I ended the prompt with JSON. we added stop seq and extra slot rules and validity rose to 100%, I locked prompt before long run.

No HTTPS from WSL)

Downloads huggingface & Github timedout, but Docker worked.

A clean-clone deploy that failed - when I tested make up from a fresh clone, seaweedFS took over a min to pull its image & script timedout before importing my images. I reordered the script & lengthed the timeout, and the 2nd attempt worked.

3) Findings)

1) Under load one pod hit it's 1-core CPU limit using 550 MiB of memory, well under 1 GiB.

2) Both pods share 1 laptop, so a second replica adds no real capacity. The autoscaling shows mechanism works. that throughput improves.

4) What I could do :-

I makes such a optimization for beaches prompt. by arranging their validation commands. A seperate development set would have been cleaner. The cluster runs on one node, so it shows nothing about fault tolerence across machines.

I have to test whole deploy from a clean clone instead only at end.

5) Use of AI)

I used an AI asst for steps guidance. All the o/p was carefully scrutinized by myself and reported only measured numbers. When results were wrong, such as failed rolling update. I kept all evidence and descriptn in AI-use.md.